

Question 1

Part (a) needs to be correct for part (b) to be marked.

Part (c) needs to be correct for part (d) to be marked.

Q1(a)

Date: _____

AB \ CD	00	01	11	10
00		0	0	0
01	0			
11	0			
10		0		

Q1(b) $(B' + C + D)(B + C + D')(A + B + C)$

Q1(c)

AB \ CD	00	01	11	10
00	1			
01		1	1	1
11		1	1	1
10	1		1	1

$B'C'D' + BD + BC + AC$

Q1(d) $B'C'D' + BD + BC + AC$

Question 2

CD \ AB	00	01	11	10
00	1 0	0 1	1 3	1 2
01	0 4	1 5	1 7	0 6
11	0 12	1 13	1 15	0 14
10	1 8	1 9	1 11	1 10

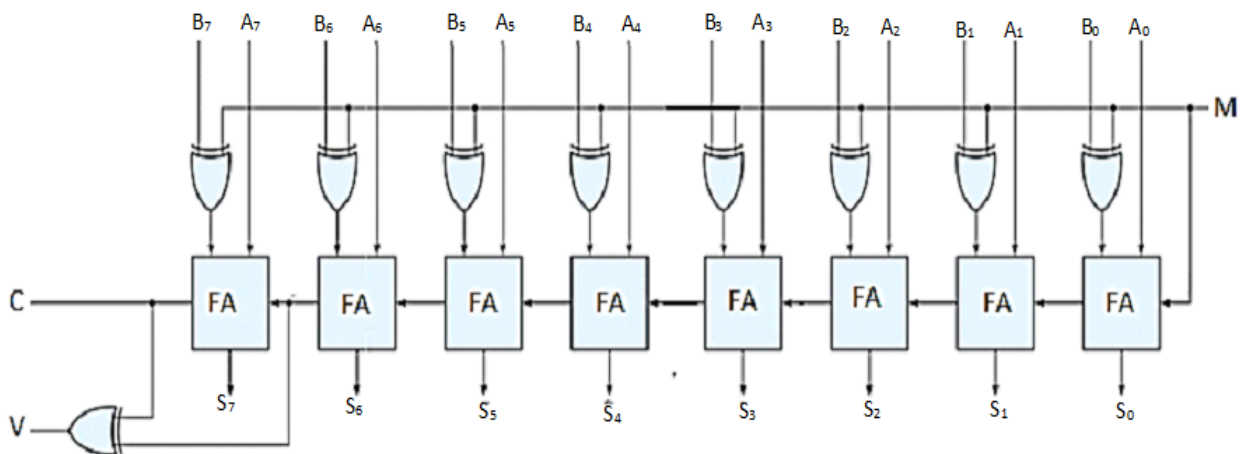
Simplified Boolean Expression

$$AD + B'D' + BD + CD$$

EPI: B'D', BD

Q3:

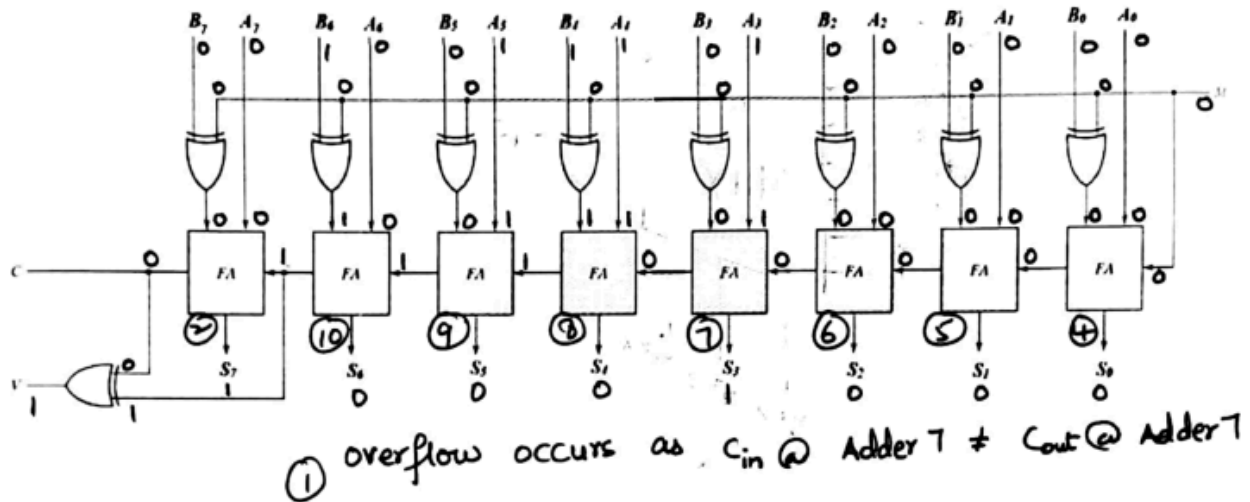
- a) [4 Marks] Solve $A+B$ in binary number system using following circuit. Where A is +56 and B is +80. Indicate if overflow occurs or not. Solve and mention each input and output at every point necessary to complete the flow of the circuit.



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the flow of the circuit.

$A = 00111000$ $B = 01010000$



- b) [3+3+2=8 Marks]** Design of a Combinational circuit to produce the 2's complement of a 4-bit binary number. (take A,B,C,D as input and E, F,G and H as output)

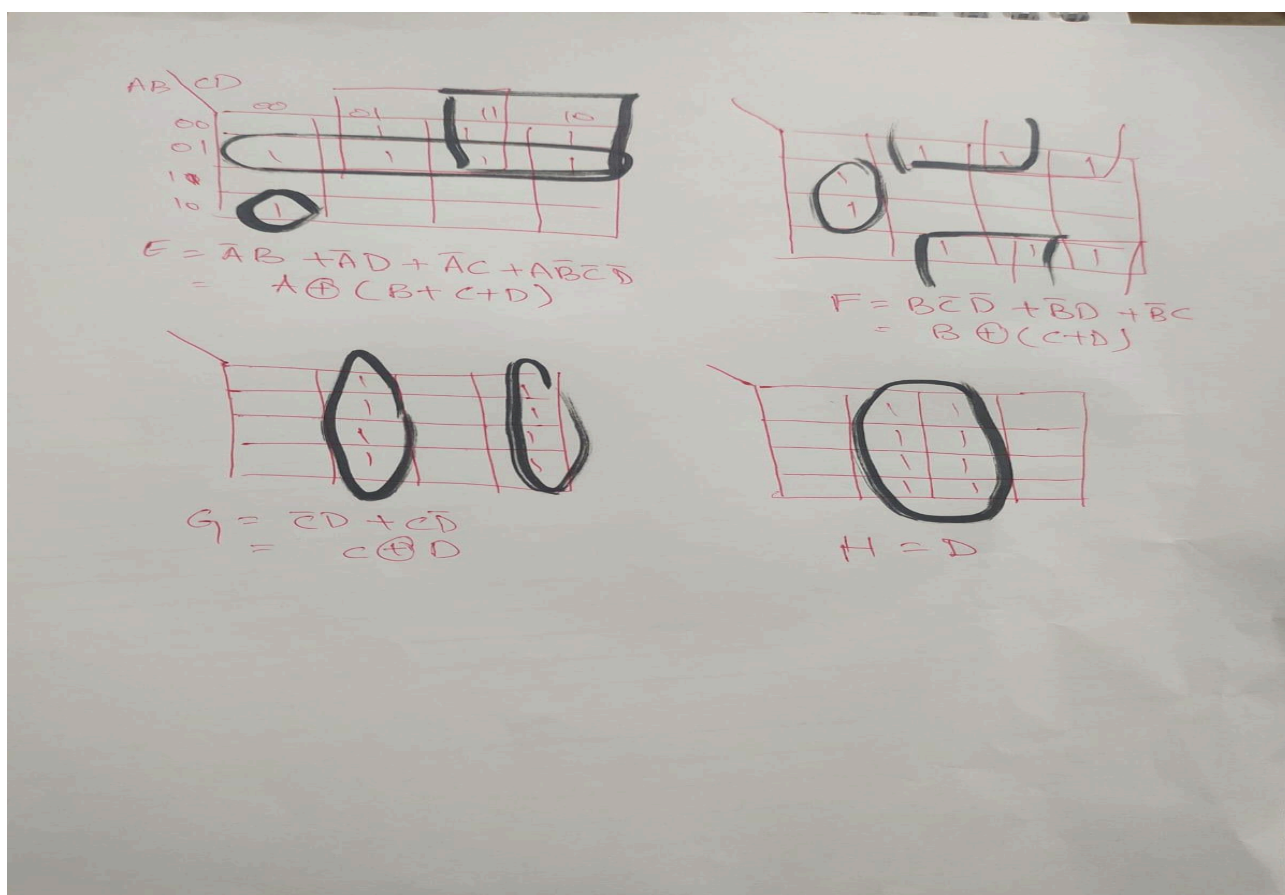
Truth Table:

Input				Output			
A	B	C	D	E	F	G	H
0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	1
0	0	1	0	1	1	1	0
0	0	1	1	1	1	0	1
0	1	0	0	1	1	0	0
0	1	0	1	1	0	1	1
0	1	1	0	1	0	1	0
0	1	1	1	1	0	0	1
1	0	0	0	1	0	0	0
1	0	0	1	0	1	1	1
1	0	1	0	0	1	1	0
1	0	1	1	0	1	0	1
1	1	0	0	0	1	0	0
1	1	0	1	0	0	1	1
1	1	1	0	0	0	1	0
1	1	1	1	0	0	0	1

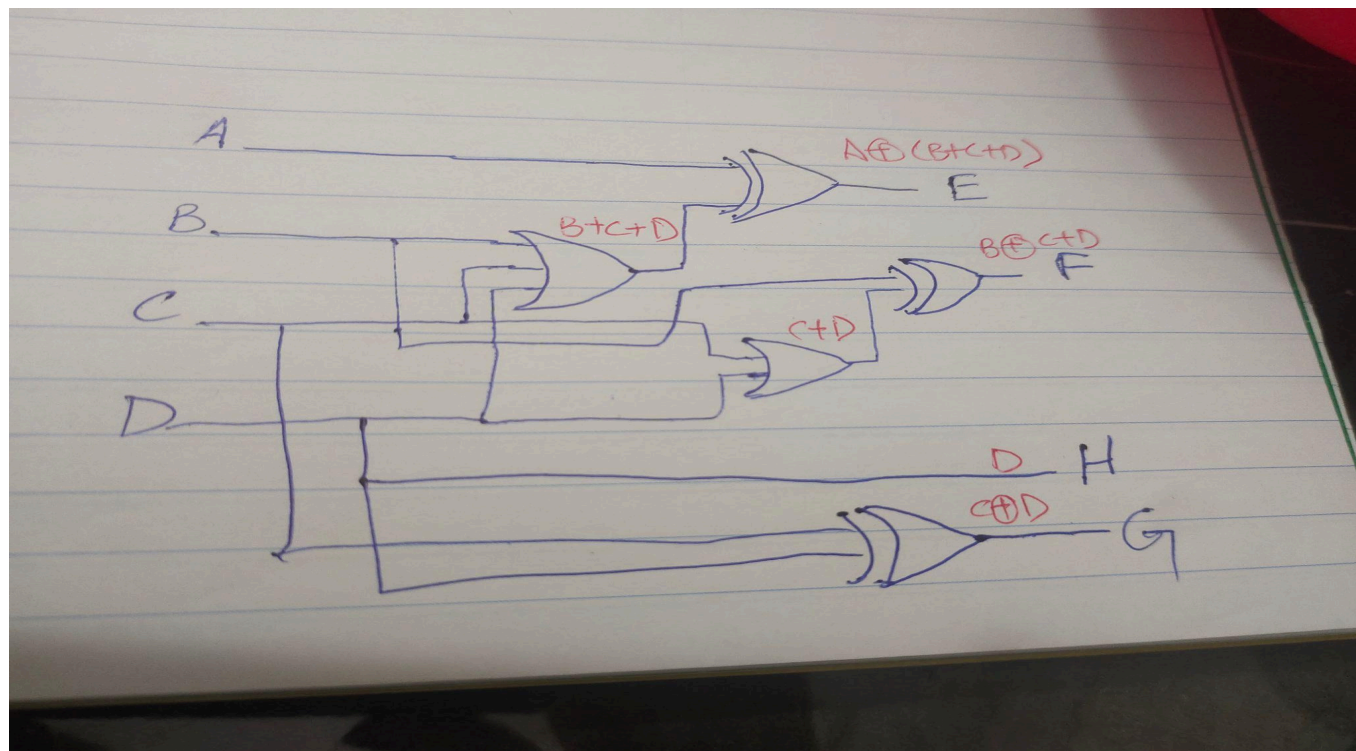
Simplification:

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Circuit Diagram:



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Q4:[2+2+2+2=8 Marks] Carry Look-Ahead Adder (CLA) Computation

Given a 4-bit CLA with inputs:

$$A=1101, B=1011, C_0=0$$

Task:

1. Compute 'Generate Gi' and 'Propagate Pi' for each bit

$$P_i = A_i \oplus B_i = 0110$$

$$G_i = A_i \cdot B_i = 1001$$

2. Calculate carries: C_1, C_2, C_3, C_4 using CLA equations

$$C_1 = G_0 + P_0 C_0$$

$$C_2 = G_1 + P_1 C_1 = G_1 + P_1(G_0 + P_0 C_0) = G_1 + P_1 G_0 + P_1 P_0 C_0$$

$$C_3 = G_2 + P_2 C_2 = G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2 P_1 P_0 C_0$$

$$C_4 = G_3 + P_3 C_3 = G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 G_0 + P_3 P_2 P_1 P_0 C_0$$

$$C_1=1, C_2=1, C_3=1, C_4=1$$

3. Determine all sum outputs

$$S_i = A_i \oplus B_i \oplus C_i \quad \rightarrow \quad S = 1000 \quad C_4 = 1$$

4. If each logic level = 2 ns, compute total CLA delay

$$\text{Given: } 1 \text{ level} = 2 \text{ ns}$$

CLA Delay: Carry logic ≈ 2 levels

Sum logic ≈ 1 level

RCA Delay (for comparison): $4 \text{ stages} \times 2 = 8 \text{ ns}$

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Q5: a) [6 Marks] Below are the two 2:4 decoders with enable, add connections, one Not gate and two OR gates to implement the given functions. Take C as Enable input and Not gate with second decoder.

A	B	C	F1	F2
0	0	0	1	0
0	0	1	0	1
0	1	0	0	0
0	1	1	1	1
1	0	0	1	0
1	0	1	1	0
1	1	0	0	1
1	1	1	1	0

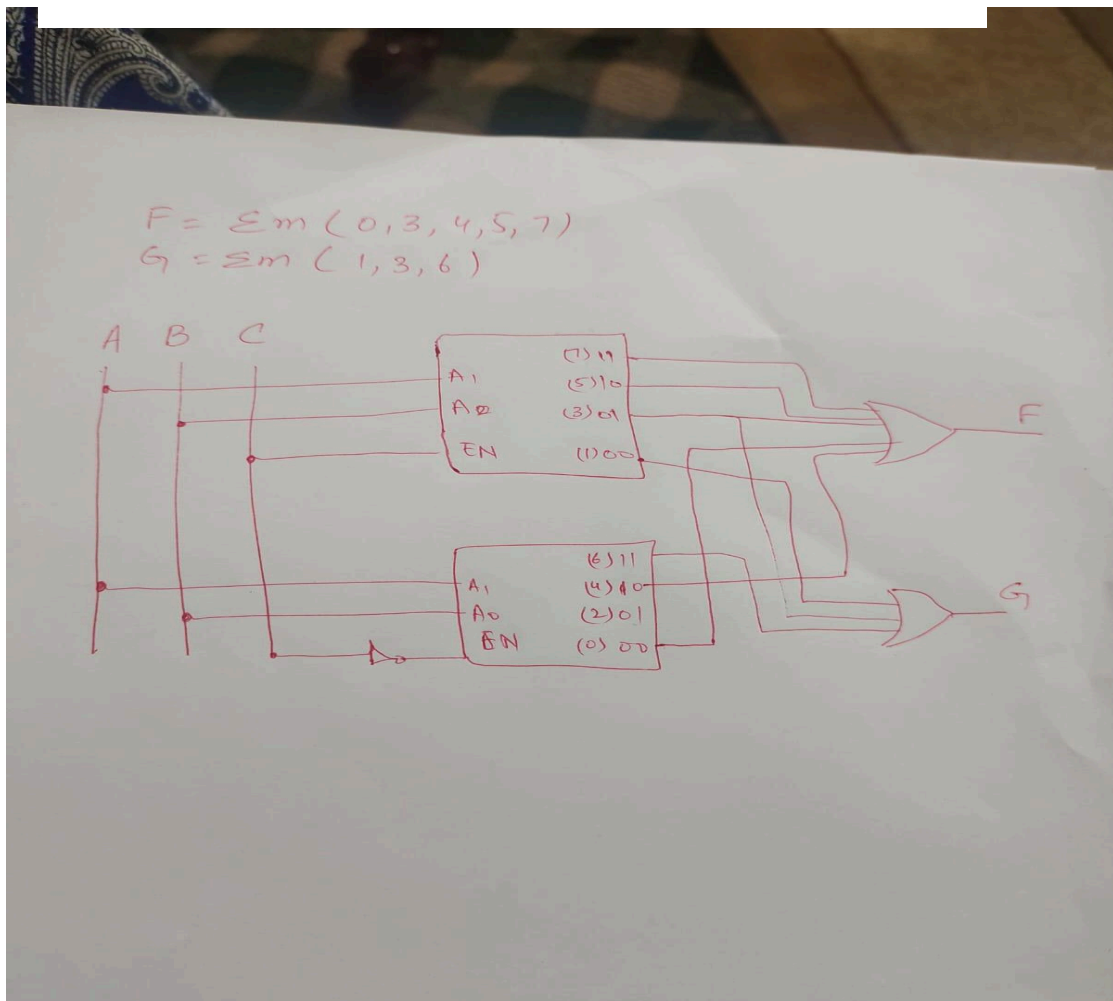
A B C

2:4 Decoder with Enable	
	11
A1	10
A0	01
EN	00

F1

2:4 Decoder with Enable	
	11
A1	10
A0	01
EN	00

F2



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- a) [4 marks] Complete the following truth table for a four-input priority encoder with following priority $D2 > D0 > D3 > D1$

D0	D1	D2	D3	A1	A0	V
0	0	0	0	x	x	0
1	x	0	x	0	0	1
0	1	0	0	0	1	1
x	x	1	x	1	0	1
0	x	0	1	1	1	1

F1